

Employee Attrition Prediction

Presented by

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Business Problem

Most employees stay, only a small portion leave. The data is highly imbalanced.

Identifying those at risk of leaving is key to effective retention and cost savings.

The imbalance makes it difficult for models to detect leavers accurately.

Improving recall on the minority class is the core challenge.

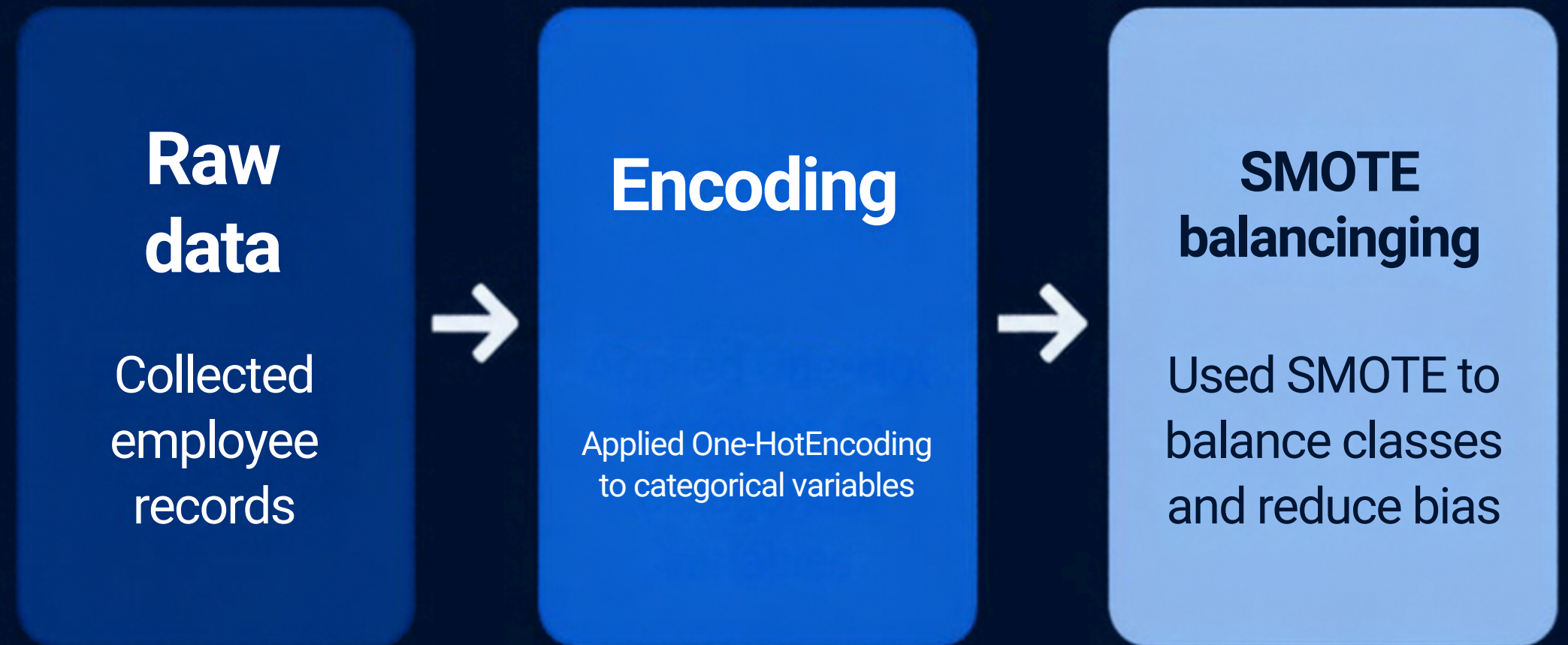


Data Preparation

We cleaned and transformed raw employee data to build a reliable model.

Categorical variables were encoded for machine readability.

We balanced the dataset to improve detection of minority attrition cases.



Key Insights & EDA

Exploratory analysis reveals the strongest drivers of employee attrition.

OVERTIME WORK FREQUENCY

Consistently associated with higher attrition due to burnout and reduced work-life balance.

LOW MONTHLY INCOME

Lower pay relative to market increases the likelihood of employees seeking other opportunities.

FREQUENT BUSINESS TRAVEL

Extensive travel correlates with job dissatisfaction and higher intent to leave.



Model & Evaluation

We selected LogisticRegression over SVM.

It delivers higher Recall, which is essential to reduce missed leavers..

Prioritizing Recall helps identify employees most likely to leave.

Recall Comparison

LogisticRegression

62%

Recall

SVM

~0%

Recall

Deployment & Impact

Deployed as an interactive
Streamlit web application

Empowers HR teams with
data-driven retention decisions

Helps reduce hiring and training costs
by addressing attrition risks early

Turning insights into action.

**Continuously improving, continually
retaining.**

